

AJAY VENKATESH

Brooklyn, NY

+1 516 585 9013

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Education

New York University | Master of Science in Computer Engineering (GPA – 3.9/4)

Sep 2024 – May 2026

• Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Skills

Programming Languages: Python, Java, C++, TypeScript, JavaScript (ES6), SQL, Scala, Swift

Web Technologies: REST API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize

Database: RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse, Oracle

Cloud: AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs)

Tools: Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Power Automate, Power BI, QuickSight, D3, Apache

AI/ML: PyTorch, Hugging Face, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, SVM, Random Forest, MLflow, PySpark

Experience

Campus Mesh

Jan 2026 – Present

Software Engineer | Real-time microservices, recommendation engine, and LLM-based AI retrieval pipelines

– Developed and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.

– Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing user engagement by **28%** and retention by **50%**.

– Authored **LLM-based retrieval pipelines** for CampusIQ (AI Assistant), improving contextual accuracy by **30%** through semantic retrieval and structured data indexing.

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern | Serverless AWS infrastructure, React 18 frontend, and Java backend systems

– Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the **system design** for approval and established Role Based Access (**4+ roles**) using **OAuth, Midway SSO, and HELIS**.

– Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD deployments**.

– Built a **React 18** frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by **40%**.

– Engineered backend services in **Java** using **Coral, Smithy, Dagger, and CBOR serialization**; **applied low-level design and design patterns** reduced backend execution time by **35%** and ensure reliability by unit and integration testing using **Mockito**.

New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

Software Developer (On-Campus) | NSF-funded AI health platform, ETL pipelines, and full-stack insurance analytics

– Architected an **NSF-funded (\$100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models for health metrix analysis and process food images.

– Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.

– Developed an insurance analytics platform (**React, Node.js, REST APIs**); deployed (**Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **\$50K+** revenue.

– Streamlined underwriting via broker portal integrated with insurer platform; implemented **document and media processing** and **WebSocket-based real-time communication**; secured **6 insurance clients**.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer | Cloud-native backend, NLP/OCR automation, and data engineering on Azure

– Developed a distributed backend on **Microsoft Azure** for a health platform, integrating with **Microsoft CRM** and implementing event-driven processing and fault tolerance.

– Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.

– Deployed an **NLP based OCR neural model** on Azure to automate structured data extraction, reducing data processing time by **75%**.

– Optimized data ingestion with **parallel processing** for **100K-row** from Excel; reduced execution time from **5 hours to 1 hour**.

– Built **Power BI dashboards** processing large datasets from multiple databases, improving reporting efficiency by **40%**.

Projects

Personal Finance Analyzer (Python Data Analysis Project) | Python, Pandas, NumPy, Matplotlib, Seaborn

– Built a data-driven application to analyze personal financial transactions and categorize expenses using **Pandas** for efficient data processing.

– Generated interactive visualizations and spending insights using **Matplotlib** and **Seaborn** to identify trends and optimize budgeting.

– Designed reusable and modular code structure following **OOP principles** to improve maintainability and scalability.

Certifications

AWS Cloud Practitioner

Azure Data Fundamentals

Microsoft Solution Architect Expert

Power BI Data Analyst Associate

Microsoft Developer Associate

Stanford Machine Learning (Coursera)